

Generalization



- **Generalization** is the process of extracting common properties from a set of entities and create a generalized entity from it.
- **Generalization is a “bottom-up approach”** in which two or more entities can be combined to form a higher level entity if they have some attributes in common.
 - ▣ subclasses are combined to make a superclass.
- **Generalization is used** to emphasize the similarities among lower-level entity set and to hide differences in the schema



Example: Generalization



- Consider we have 3 sub entities Car, Bus and Motorcycle. Now these three entities can be generalized into one higher-level entity (or super class) named as **Vehicle**.

Superclass

Vehicle

ISA

Car

Bus

Motorcycle

Sub-classes

Generalization
(Bottom-up
Approach)



Specialization



- Specialization is **opposite** of Generalization
- In **Specialization**, an entity is broken down into sub-entities based on their characteristics.
- **Specialization is a “Top-down approach”** where higher level entity is specialized into two or more lower level entities.
- **Specialization is used** to identify the subset of an entity set that shares some distinguishing characteristics.
- **Specialization** can be repeatedly applied to refine the design of schema
- depicted by triangle component labeled **ISA**



Example: Specialization



□ **Vehicle** entity can be a Car, Truck or Motorcycle.

- Normally, the superclass is defined first, the subclass and its related attributes are defined next, and relationship set are then added.

Superclass

Vehicle ✓



Sub-classes

Car

Bus

Motorcycle

Specialization
(Top-down
Approach)

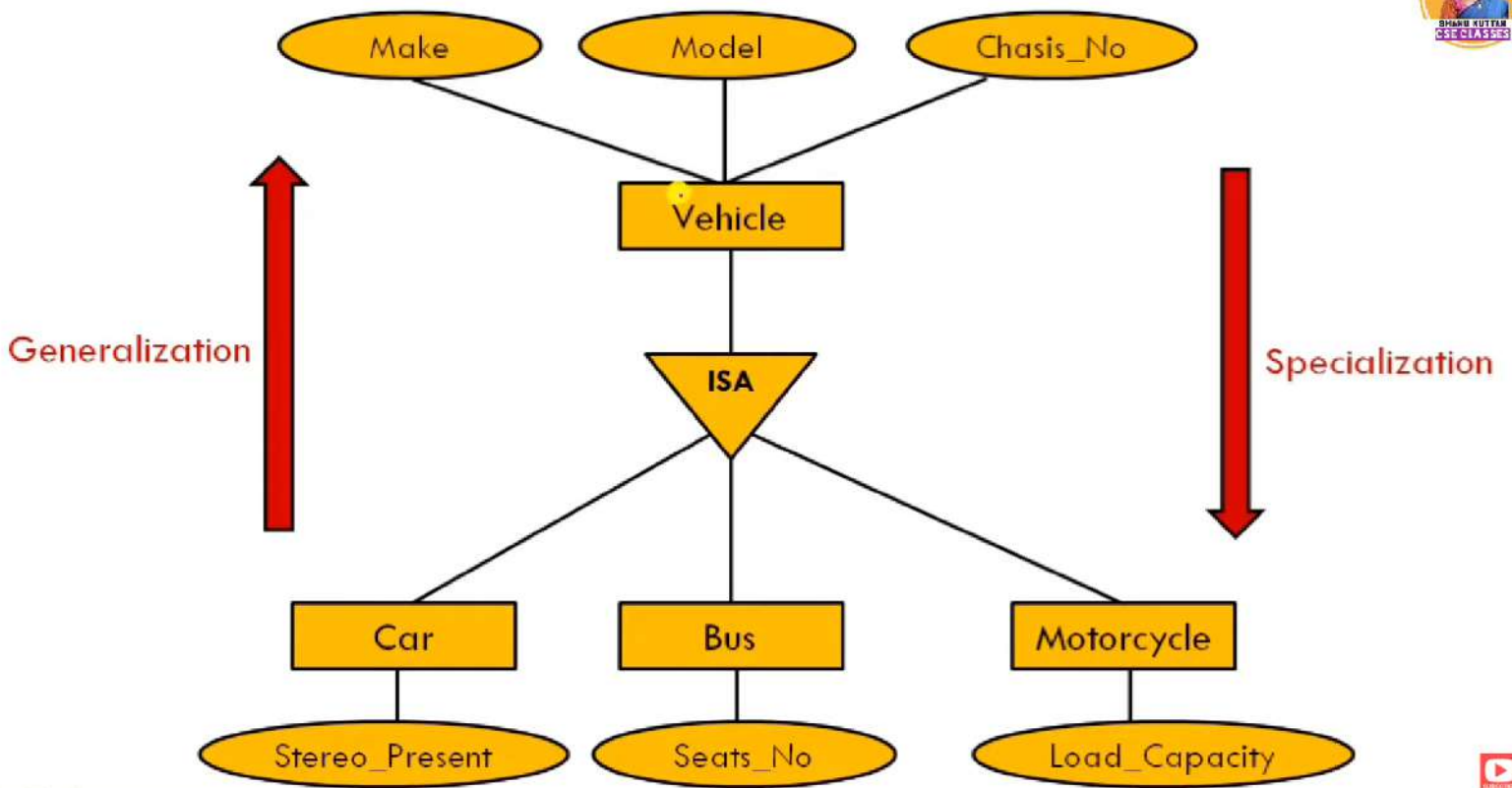


Inheritance



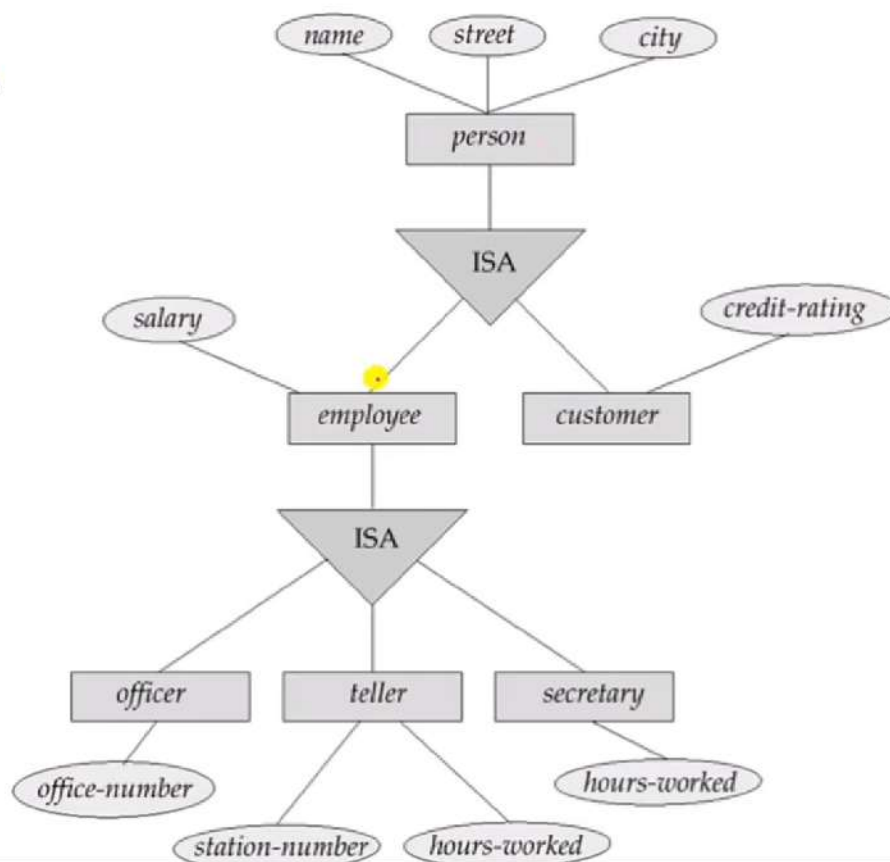
- **Inheritance** is an important feature of generalization and specialization.
- **Attribute inheritance** allows lower level entities to inherit the attributes of higher level entities.
 - **For example**, Consider relations Car and Bus inheriting the attributes of Vehicle. Thus, Car is described by attributes of super-class Vehicle as well as its own attributes.
- This also extends to **Participation Inheritance** in which relationships involving higher-level entity-sets are also inherited by lower-level entity-sets.
 - A lower-level entity-set can participate in its own relationship-sets, too



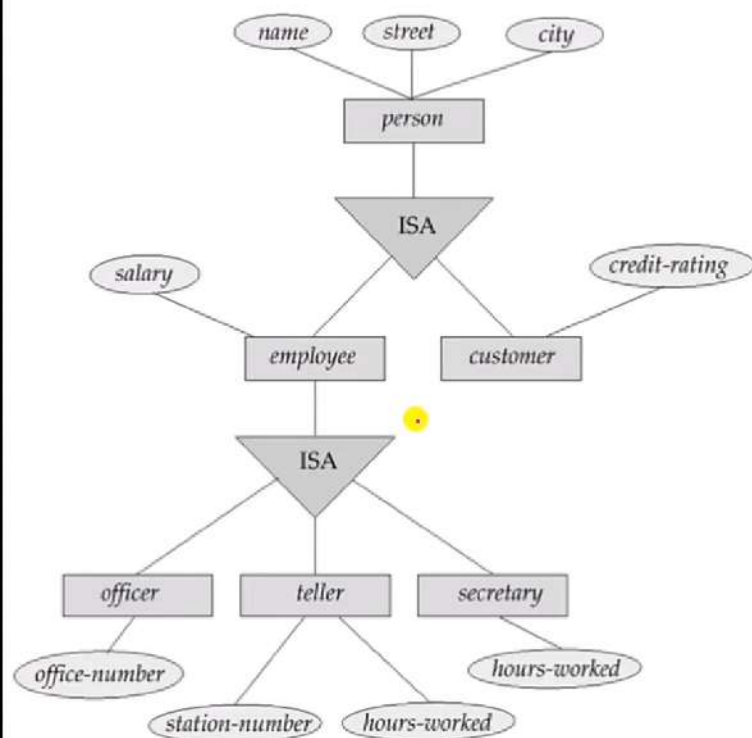


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Specialization



How Schema or Tables can be formed?



Four tables can be formed:

1. **customer** (name, street, city, credit_rating)
2. **officer** (name, street, city, salary, office_number)
3. **teller** (name, street, city, salary, station_number, hours_worked)
4. **secretary** (name, street, city, salary, hours_worked)

